

Images (columns)

	I_1	I_2	I_3	I_4
T_1	s_{11}	s_{12}	s_{13}	s_{14}
T_2	s_{21}	s_{22}	s_{23}	s_{24}
T_3	s_{31}	s_{32}	s_{33}	s_{34}
T_4	s_{41}	s_{42}	s_{43}	s_{44}

image → text: softmax over each column → CE, target = i (matching text)

text → image: softmax over each row → CE, target = i (matching image)

target = diagonal

text → image: softmax over each row → CE, target = i (matching image)

Similarity matrix S ,
 $s_{ij} = \cos(\text{text}_i, \text{image}_j)$

$$\mathcal{L} = \frac{1}{2} \left(\mathcal{L}_{\text{img} \rightarrow \text{txt}} + \mathcal{L}_{\text{txt} \rightarrow \text{img}} \right)$$

$$\mathcal{L}_{\text{txt} \rightarrow \text{img}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^N \exp(s_{ij}/\tau)}$$

τ = temperature (logit scale)